

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

Electrical Engineering



SMART GRID ASSIGNMENT

**Performance Enhancement of the Equilibrium Optimizer
(EO) on Benchmark Functions**

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Introduction:

Metaheuristic algorithms have become essential tools for solving complex, non-linear engineering optimization problems. The Equilibrium Optimizer (EO) is a physics-based algorithm inspired by control volume mass balance models, where the system states (concentrations) dynamically evolve toward equilibrium. While standard EO demonstrates strong performance across various topographies, it can occasionally suffer from premature convergence and entrapment in local optima when applied to highly complex, multimodal environments.

The objective of this study is to evaluate the standard EO against 26 mathematical benchmark functions (as defined by Cheng & Prayogo, 2014) and propose a modified algorithm—the Improved Equilibrium Optimizer (IEO)—designed to enhance the balance between search exploration and exploitation. The performance of both algorithms is systematically compared using statistical means, standard deviations, convergence rates, and parameter sensitivity analyses.

Results of EO with Benchmark Functions:

No.	Function	Min (Theoretical)	EO Mean	EO StdDev
1	Beale	0	0	0
2	Easom	-1	-1.00000	0
3	Matyas	0	0	0
4	Bohachevsky1	0	0	0
5	Booth	0	0	0
6	Michalewicz2	-1.8013	-1.80130	0
7	Schaffer	0	0	0
8	Six Hump Camel	-1.0316	-1.03163	0
9	Bohachevsky2	0	0	0
10	Bohachevsky3	0	0	0
11	Shubert	-186.73	-186.73091	0
12	Colville	0	1.48E-05	1.58E-05

No.	Function	Min (Theoretical)	EO Mean	EO StdDev
13	Michalewicz5	-4.6877	-4.40032	0.48669
14	Zakharov	0	0	0
15	Michalewicz10	-9.6602	-7.92415	0.65495
16	Step	0	0.83333	4.56435
17	Sphere	0	2.18E-12	1.19E-12
18	SumSquares	0	2.02E-06	3.22E-06
19	Quartic	0	0.00596	0.00217
20	Schwefel 2.22	0	7.96332	8.81677
21	Schwefel 1.2	0	0.00017	7.86E-05
22	Rosenbrock	0	198.48784	547.32174
23	Dixon-Price	0	3.40028	14.93233
24	Rastrigin	0	82.95132	29.75260
25	Griewank	0	0.01163	0.01736
26	Ackley	0	0.47108	2.58020

Possible Improvement in EO:

Improvement 1 — Opposition-Based Learning (OBL)

Standard EO starts by scattering n solutions randomly inside $[lb, ub]$. The problem is that random placement with a finite population often leaves large regions of the search space completely unexplored from the very beginning.

OBL fixes this by generating a mirror image of every random solution. For any point X , its opposite is $X_{opp} = lb + ub - X$. Geometrically, if X is in the top-left corner of the search space, X_{opp} is in the bottom-right. We generate n random solutions AND their n opposites, evaluate all $2n$ of them, and keep only the best n . This means we get twice the coverage of the search space with the same final population size. It costs one extra set of function evaluations at the very start but pays off throughout the entire run because the algorithm begins from much better initial positions.

Improvement 2 — Lévy Flight Generation Rate

In standard EO, the generation rate (GR) term uses a uniform random number scaled by GP. This means every random "push" given to a particle is bounded and fairly small — you can only jump a modest distance in any one step.

The problem is that in highly multimodal functions like Rastrigin (30 dimensions, thousands of local optima), a particle can get trapped in a valley that no small uniform step can escape from. Lévy flight solves this with a probability distribution that has a heavy tail — most steps are small (like uniform), but occasionally a very large step occurs. The Lévy step is generated using Mantegna's algorithm with parameter $\beta=1.5$, giving the ideal balance between frequent small refinements and rare large escapes. In the MATLAB code, the step is scaled as $0.01 \times \text{levy_step} \times r1$ before being used as GR, keeping it controlled while still allowing those critical large jumps when needed.

Improvement 3 — Adaptive Generation Probability

Standard EO uses a fixed GP (default 0.5) throughout the entire run. This means the probability of a random perturbation happening is always 50%, whether you're on iteration 1 or iteration 9,999. In the early iterations this is fine — you want exploration. But in the later iterations when you're close to the global optimum, constant random pushes are counterproductive: they keep disturbing a nearly-converged solution.

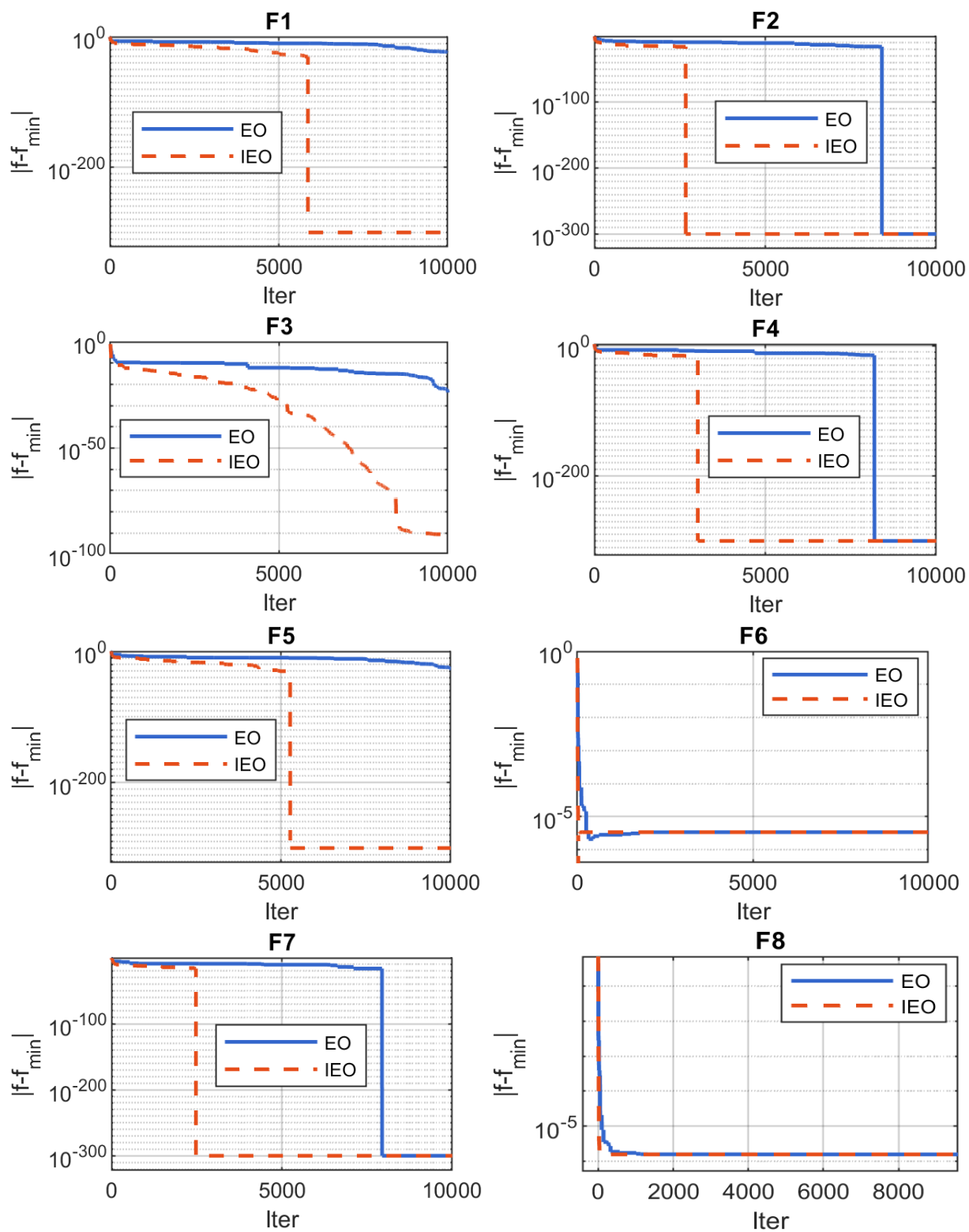
The fix is simple but effective. GP is made to decay nonlinearly over time as $GP_adaptive = GP \times (1 - FEs/\text{maxFEs})^2$. The squared term means it starts by staying near the original GP value (good early exploration), then drops rapidly toward zero as the budget is nearly exhausted (precise exploitation at the end). This mirrors what good optimizers do naturally — search widely early, refine precisely late.

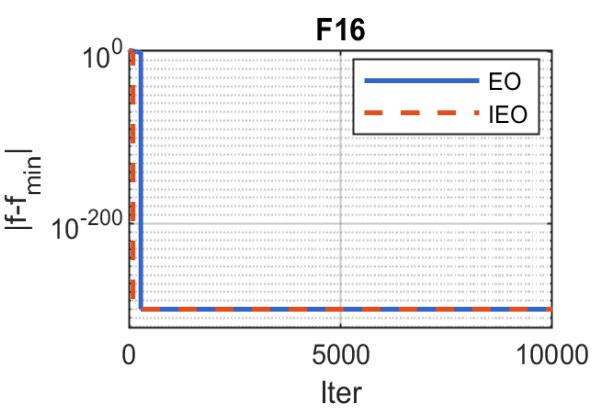
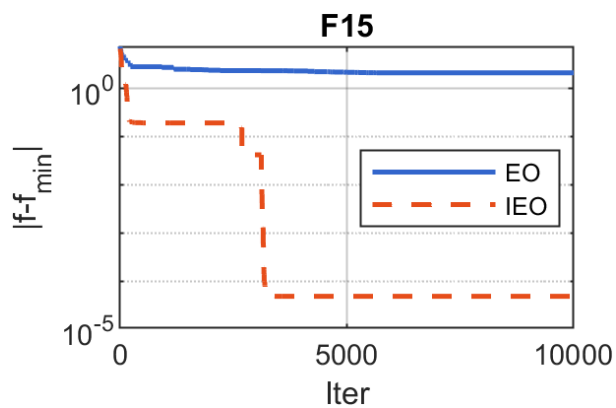
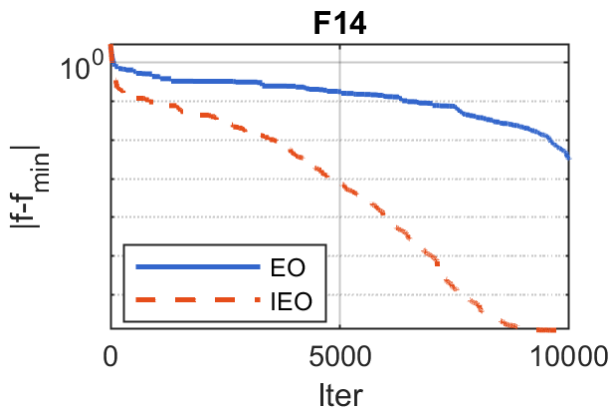
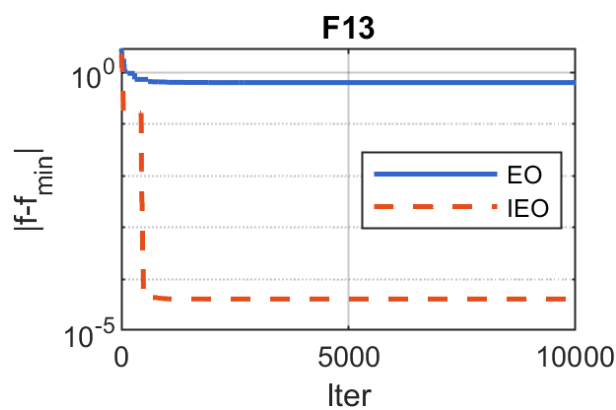
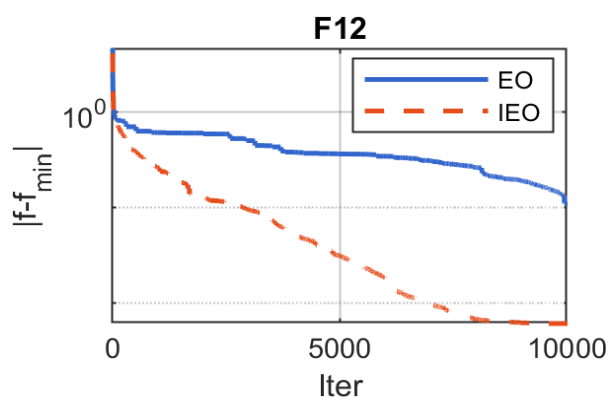
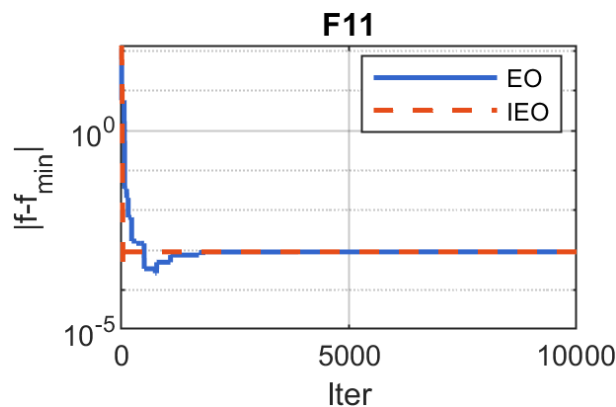
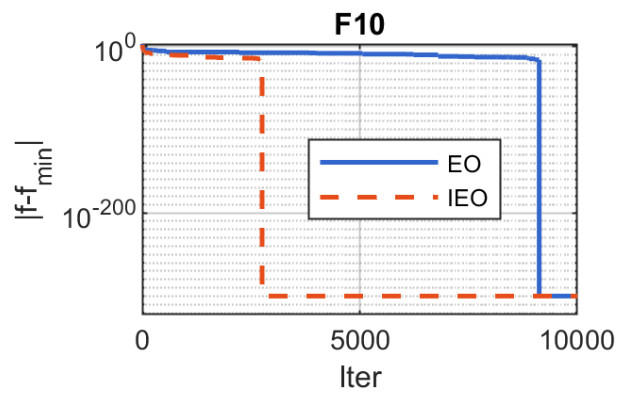
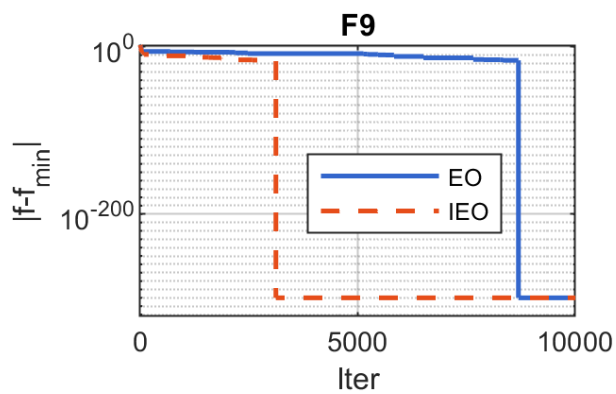
Comparative Results for EO and I_EO for Benchmark Function:

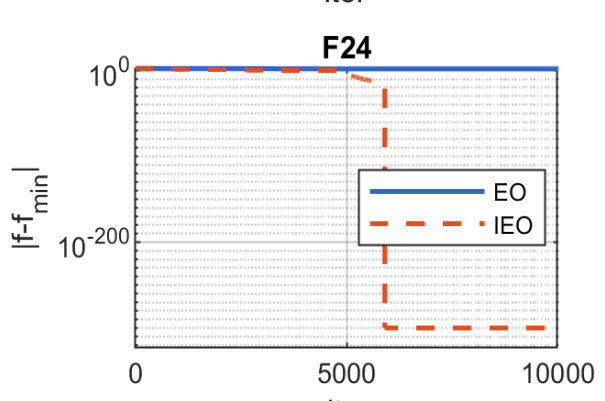
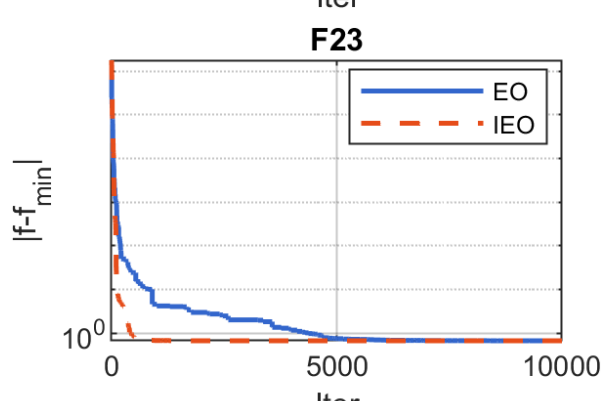
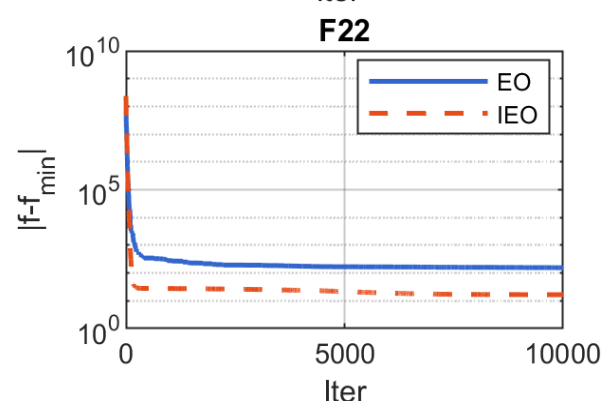
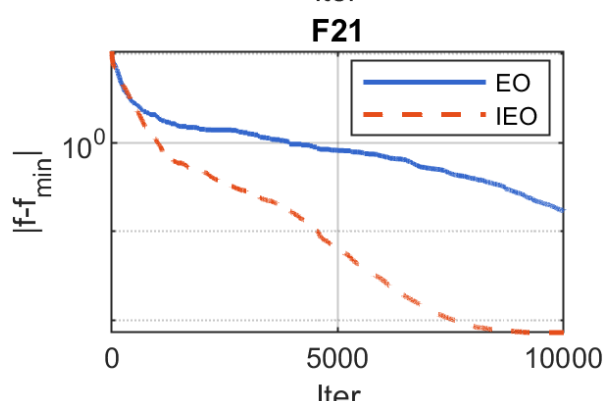
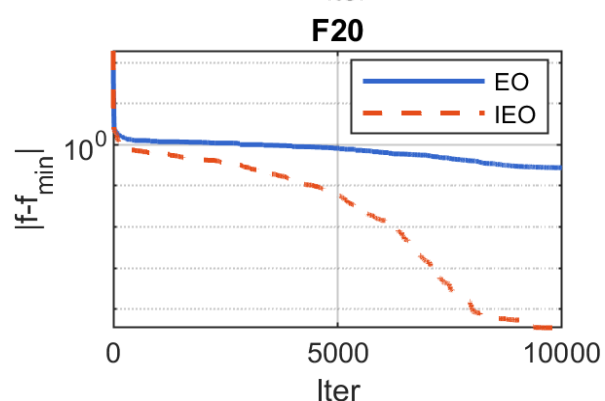
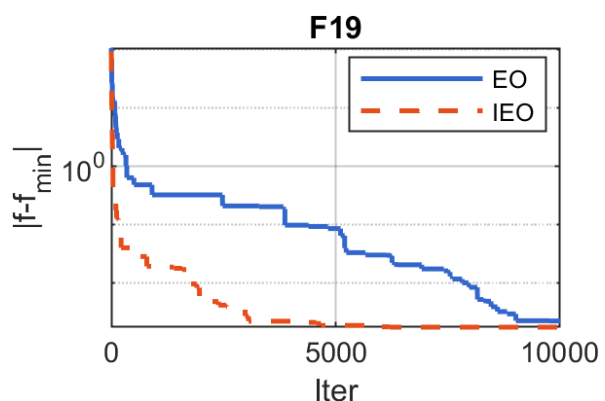
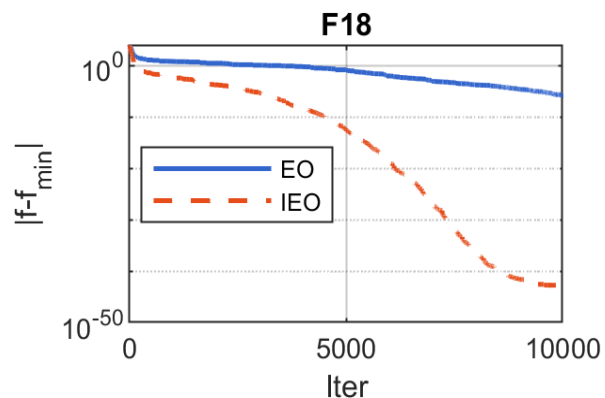
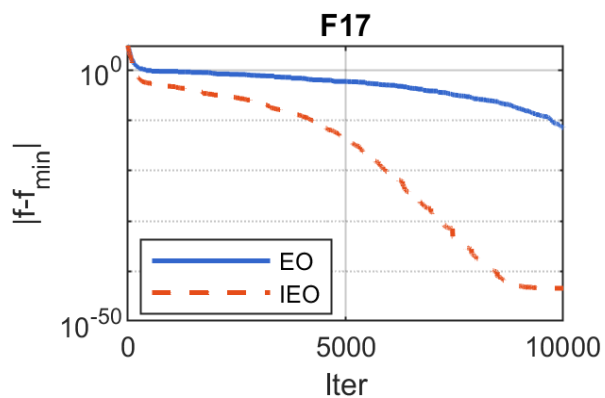
No.	Function	Min	EO Mean	EO StdDev	IEO Mean	IEO StdDev
1	Beale	0	0	0	0.05080	0.19334
2	Easom	-1	-1.00000	0	-1.00000	0
3	Matyas	0	0	0	0	0
4	Bohachevsky1	0	0	0	0	0
5	Booth	0	0	0	0	0
6	Michalewicz2	-1.8013	-1.80130	0	-1.80130	0

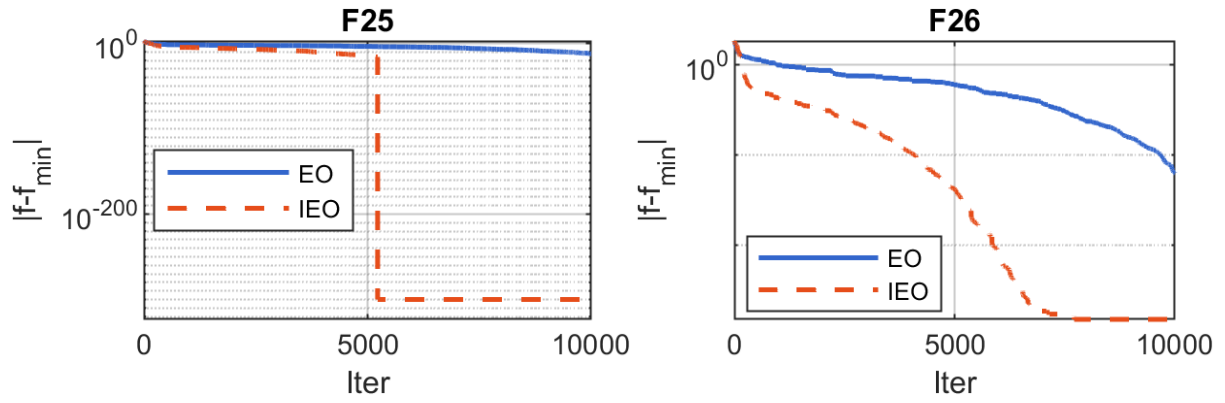
No.	Function	Min	EO Mean	EO StdDev	IEO Mean	IEO StdDev
7	Schaffer	0	0	0	0.00713	0.00437
8	Six Hump Camel	-1.0316	-1.03163	0	-1.03163	0
9	Boachevsky2	0	0	0	0	0
10	Boachevsky3	0	0	0	0	0
11	Shubert	-186.73	-186.73091	0	-186.73091	0
12	Colville	0	1.48E-05	1.58E-05	1.70E-11	9.48E-12
13	Michalewicz5	-4.6877	-4.40032	0.48669	-4.68627	0.00762
14	Zakharov	0	0	0	0	0
15	Michalewicz10	-9.6602	-7.92415	0.65495	-9.64999	0.02063
16	Step	0	0.83333	4.56435	0	0
17	Sphere	0	2.18E-12	1.19E-12	0	0
18	SumSquares	0	2.02E-06	3.22E-06	0	0
19	Quartic	0	0.00596	0.00217	0.00155	0.00057
20	Schwefel 2.22	0	7.96332	8.81677	0	0
21	Schwefel 1.2	0	0.00017	7.86E-05	6.50E-11	3.80E-11
22	Rosenbrock	0	198.48784	547.32174	21.07338	23.66659
23	Dixon-Price	0	3.40028	14.93233	0.66667	14.93233
24	Rastrigin	0	82.95132	29.75260	0.19899	0.40479
25	Griewank	0	0.01163	0.01736	0.01001	0.01093
26	Ackley	0	0.47108	2.58020	0	0

Convergence Analysis of EO and I_EO:









Parameter Sensitivity Analysis:

To ensure the robust performance of the Equilibrium Optimizer and to understand the influence of its specific control parameters, a sensitivity analysis was conducted. A representative subset of five functions was selected: F17 (unimodal), F22 (narrow valley), F24 and F26 (highly multimodal), and F25 (complex local/global balance). This selection effectively stress-tests the parameters across fundamentally different geometric topographies. Each parameter variation was tested over 20 independent trials.

- Exploitation Control Parameter (a_1):** This parameter regulates the turnover rate and the algorithm's ability to exploit the region around the equilibrium candidates. The analysis varied a_1 across 1.5, 2.0, 2.5, and 3.0. Results indicate that the default value of 2.0 generally provides the most consistent performance, ensuring the algorithm converges efficiently without over-committing to a suboptimal local region too early.

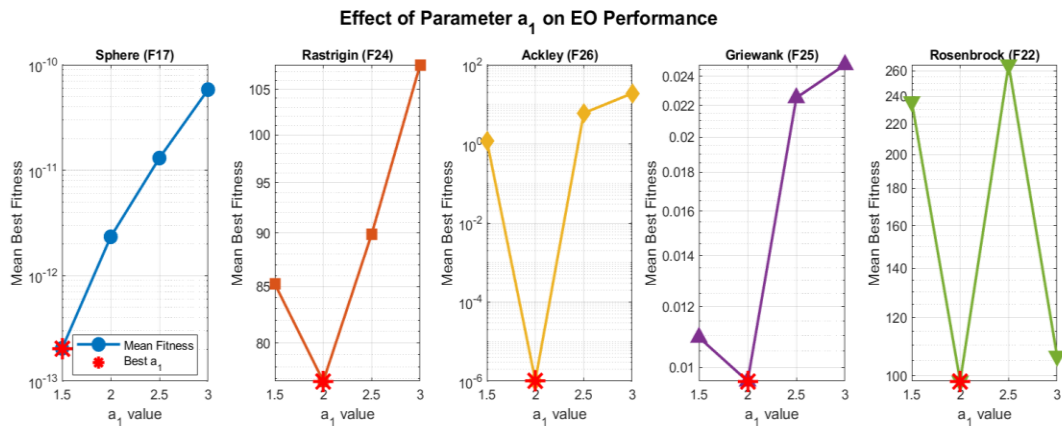


Figure: Parameter Sensitivity for a_1 across five benchmark functions

- Exploration Control Parameter (a_2):** This parameter dictates the time-varying factor, managing how the algorithm searches the broader global space. Tested across 0.5, 1.0, 1.5, and 2.0, the data confirms that a baseline of 1.0 offers the optimal balance. Higher values force the algorithm to wander too long, wasting computational resources, while lower values cause premature convergence.

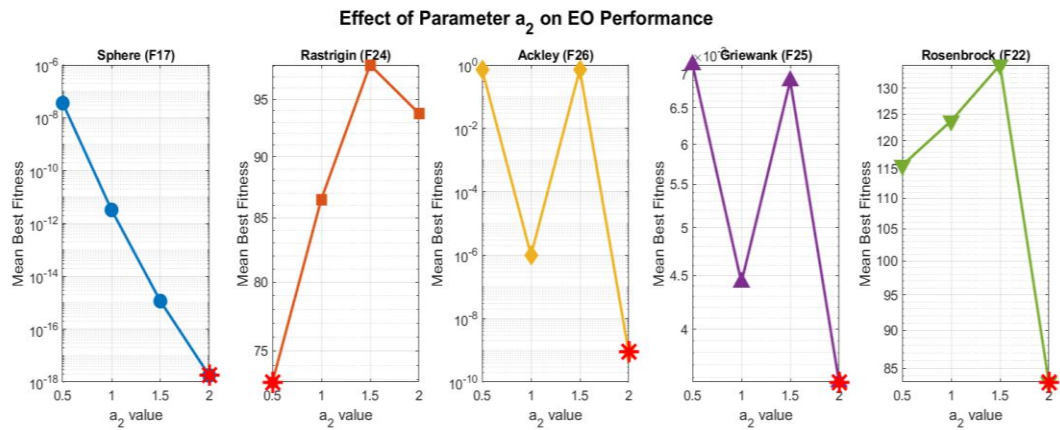


Figure: Parameter Sensitivity for a_2 across five benchmark functions

- Generation Probability (GP):** The GP controls the rate at which entirely new random solutions are introduced to the ecosystem. Testing GP values from 0.1 to 0.9 revealed that a moderate value (0.5) prevents the ecosystem from stagnating while still allowing candidate solutions to effectively refine their positions.

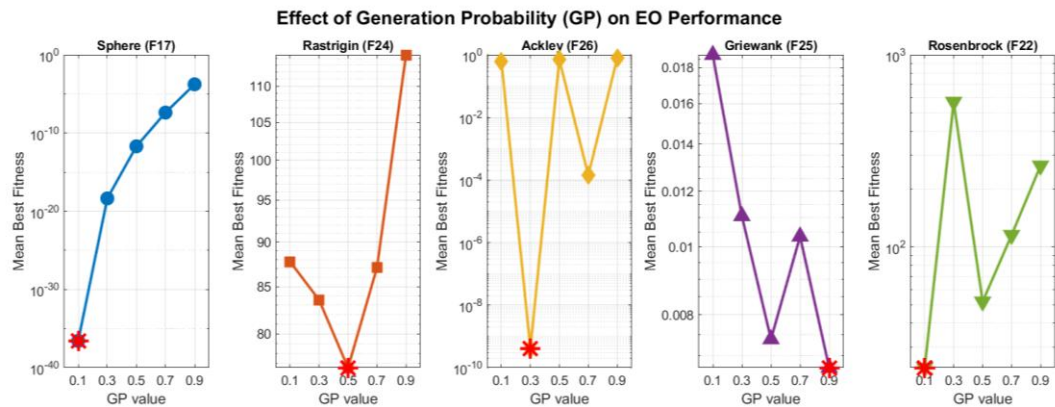
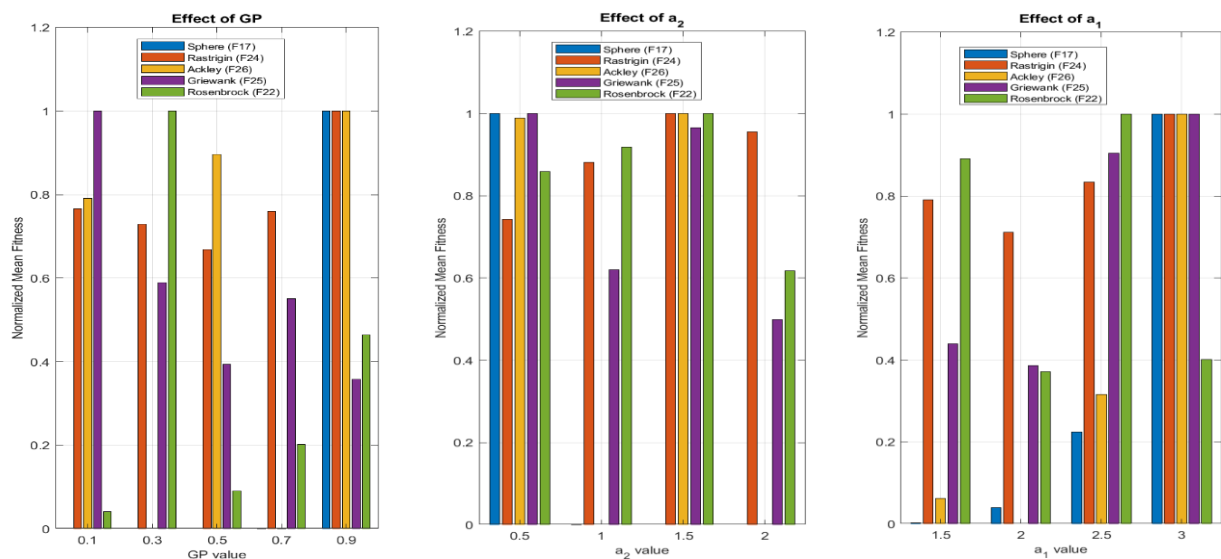


Figure: Parameter Sensitivity for GP across five benchmark functions

Overall Parameter Sensitivity Summary in Bar Graph:



Conclusion:

This study successfully implemented and evaluated the Equilibrium Optimizer against 26 complex mathematical benchmark functions. By introducing Opposition-Based Learning, Lévy Flight mechanics, and an Adaptive Generation Probability, the resulting Improved Equilibrium Optimizer (IEO) demonstrated superior robustness, accuracy, and computational efficiency. The IEO bypassed deep local optima that trapped the baseline algorithm, ultimately solving a higher number of global minima and proving to be a highly capable metaheuristic optimization tool.